

TechSud — Audit et sécurisation d'une infrastructure informatique

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Projet AEGIS
— Sécurité des
Systèmes
d'Information



Groupe 8 :
Quentin Godebert
— Jules Elie —
Evann Carnot



Objectifs : auditer,
sécuriser,
automatiser,
documenter

.IPSSI'

Contexte de l'incident et répartition des rôles

- Alerte CPU à 98 % hors heures ouvrées
- Connexions SSH suspectes depuis une IP inconnue
- Fichier suspect et cron anormale
- Webshell probable sur le serveur web
- Objectif : identifier les vecteurs probables et sécuriser
- Jules : VM, SSH, UFW, fail2ban
- Evann : script Python
- Quentin : rapport, captures, soutenance

Mise en place de la VM et audit initial

- Déploiement d'une VM Ubuntu Server
- Vérification réseau
- Observation des ports et services
- Scan initial depuis un poste externe

```
tech@tech:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default
    link/ether 08:00:27:3c:b7:cd brd ff:ff:ff:ff:ff:ff
    inet 10.74.2.204/22 metric 100 brd 10.74.3.255 scope global dynamic enp0s3
        valid_lft 7144sec preferred_lft 7144sec
    inet6 fe80::a00:27ff:fe3c:b7cd/64 scope link
        valid_lft forever preferred_lft forever
```



TechSud



En fonction

Interface 1: Intel PRO/1000 MT Desktop (Interface pont Intel(R) Wi-Fi 6E AX211 160MHz)

Interface 2: Intel PRO/1000 MT Desktop (NAT)

Vulnérabilités et risques prioritaires

- Exposition de SSH
- Port standard 22 au départ
- Accès root à éviter
- Absence initiale de firewall actif
- Absence initiale de fail2ban
- Logs non centralisés dans le scénario TechSud

```
tech@tech:~$ systemctl status ssh
* ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: inactive (dead)
 TriggeredBy: ● ssh.socket
      Docs: man:sshd(8)
           man:sshd_config(5)
```

```
PS C:\WINDOWS\system32> nmap -sV 10.74.2.204
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-13 12:33 +0200
Nmap scan report for 10.74.2.204
Host is up (0.018s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 9.6p1 Ubuntu 3ubuntu13.15 (Ubuntu Linux; protocol 2.0)
MAC Address: 00:72:EE:18:05:BA (Intel Corporate)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.78 seconds
```

Durcissement de l'accès administrateur

- Port SSH changé en 2222
- PermitRootLogin no
- PasswordAuthentication no
- Authentification par clé
- Accès final avec admintech

```
tech@tech:~$ sudo grep -E '^(Port|PermitRootLogin|PasswordAuthentication|PubkeyAuthentication)' /etc/ssh/sshd_config
Port 2222
PermitRootLogin no
PubkeyAuthentication yes
PasswordAuthentication no
tech@tech:~$
```

```
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-107-generic x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System information as of mar. 14 avril 2026 13:55:12 UTC

System load:  0.03               Processes:            152
Usage of /:   45.1% of 13.67GB   Users logged in:     1
Memory usage: 9%                IPv4 address for enp0s3: 10.74.2.217
Swap usage:   0%

La maintenance de sécurité étendue pour Applications n'est pas activée.

0 mise à jour peut être appliquée immédiatement.

Activez ESM Apps pour recevoir des futures mises à jour de sécurité supplémentaires.
Visitez https://ubuntu.com/esm ou exécutez : sudo pro status

*** Le système doit être redémarré ***
Last login: Tue Apr 14 13:33:42 2026 from 10.74.2.213
admintech@tech:~$
```

Réduction de l'exposition

- UFW activé
- Ouverture contrôlée de 2222/tcp
- fail2ban installé
- Jail sshd active

```
admintech@tech:~$ sudo ufw allow 2222/tcp
Rules updated
Rules updated (v6)
admintech@tech:~$ sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
admintech@tech:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To Action From
--
2222/tcp ALLOW IN Anywhere
2222/tcp (v6) ALLOW IN Anywhere (v6)
```

```
admintech@tech:~$ sudo cat /etc/fail2ban/jail.local
[sshd]
enabled = true
port = 2222
logpath = %(sshd_log)s
backend = systemd
maxretry = 3
findtime = 10m
bantime = 10m
```

Réaction aux tentatives abusives

Tests de bannissements réalisés

```
PS C:\WINDOWS\system32> ssh -p 2222 usertest1@10.74.2.224
usertest1@10.74.2.224: Permission denied (publickey).
PS C:\WINDOWS\system32> ssh -p 2222 administrateur@10.74.2.224
administrateur@10.74.2.224: Permission denied (publickey).
```

```
admintech@tech:~$ sudo fail2ban-client status sshd
[sudo] password for admintech:
Status for the jail: sshd
|- Filter
| |- Currently failed: 0
| |- Total failed: 0
| `-- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd
`- Actions
  |- Currently banned: 0
  |- Total banned: 1
  `-- Banned IP list:
admintech@tech:~$ sudo journalctl -u fail2ban --no-pager -n 50
avril 15 09:24:39 tech systemd[1]: Started fail2ban.service - Fail2Ban Service.
avril 15 09:24:40 tech fail2ban-server[2260]: 2026-04-15 09:24:40,295 fail2ban.configreader [2260]: WARNING 'allowip' not defined in 'Definition'. Using default one: 'auto'
avril 15 09:24:42 tech fail2ban-server[2260]: Server ready
avril 15 09:28:38 tech systemd[1]: Stopping fail2ban.service - Fail2Ban Service...
avril 15 09:28:39 tech fail2ban-client[3276]: Shutdown successful
avril 15 09:28:39 tech systemd[1]: fail2ban.service: Deactivated successfully.
avril 15 09:28:39 tech systemd[1]: Stopped fail2ban.service - Fail2Ban Service.
avril 15 09:28:39 tech systemd[1]: fail2ban.service: Consumed 4.107s CPU time, 28.6M memory peak, 0B memory swap peak.
avril 15 09:28:39 tech systemd[1]: Started fail2ban.service - Fail2Ban Service.
avril 15 09:28:40 tech fail2ban-server[3279]: 2026-04-15 09:28:40,330 fail2ban.configreader [3279]: WARNING 'allowip' not defined in 'Definition'. Using default one: 'auto'
avril 15 09:28:40 tech fail2ban-server[3279]: Server ready
-- Boot d936f7909314444e9909d613a477d958 --
avril 15 09:40:43 tech systemd[1]: Started fail2ban.service - Fail2Ban Service.
avril 15 09:40:44 tech fail2ban-server[1107]: 2026-04-15 09:40:44,469 fail2ban.configreader [1107]: WARNING 'allowip' not defined in 'Definition'. Using default one: 'auto'
avril 15 09:40:44 tech fail2ban-server[1107]: Server ready
```

Surveillance

- Lecture des logs SSH et fail2ban
- Vérification des connexions et événements

```
admintech@tech:~$ sudo journalctl -u fail2ban --no-pager -n 50
avril 15 09:24:39 tech systemd[1]: Started fail2ban.service - Fail2Ban Service.
avril 15 09:24:40 tech fail2ban-server[2260]: 2026-04-15 09:24:40,295 fail2ban.configreader [2260]: WARNING 'allowipv6' not defined in 'Definition'. Using default one: 'auto'
avril 15 09:24:42 tech fail2ban-server[2260]: Server ready
avril 15 09:28:38 tech systemd[1]: Stopping fail2ban.service - Fail2Ban Service...
avril 15 09:28:39 tech fail2ban-client[3276]: Shutdown successful
avril 15 09:28:39 tech systemd[1]: fail2ban.service: Deactivated successfully.
avril 15 09:28:39 tech systemd[1]: Stopped fail2ban.service - Fail2Ban Service.
avril 15 09:28:39 tech systemd[1]: fail2ban.service: Consumed 4.107s CPU time, 28.6M memory peak, 0B memory swap peak.
avril 15 09:28:39 tech systemd[1]: Started fail2ban.service - Fail2Ban Service.
avril 15 09:28:40 tech fail2ban-server[3279]: 2026-04-15 09:28:40,330 fail2ban.configreader [3279]: WARNING 'allowipv6' not defined in 'Definition'. Using default one: 'auto'
avril 15 09:28:40 tech fail2ban-server[3279]: Server ready
```

Script Python d'audit

```
admin@tech:~$ sudo python3 audit_techsud.py
=== Audit TechSud ===

Date : 2026-04-15 10:38:17
Machine : tech

--- SSH ---
Statut : active
Port configuré : 2222
PermitRootLogin : no
PasswordAuthentication : no

--- Outils de sécurité ---
UFW : actif
fail2ban installé : True
fail2ban statut : active

--- Ports ouverts ---
53, 68, 2222

--- Services actifs ---
- cron.service
- dbus.service
- fail2ban.service
- fwupd.service
- getty@tty1.service
- ModemManager.service
- multipathd.service
- polkit.service
- rsyslog.service
- ssh.service
- systemd-journald.service
- systemd-logind.service
- systemd-networkd.service
- systemd-resolved.service
- systemd-udev.service
- udisks2.service
- unattended-upgrades.service
- upower.service
- user@1000.service
- user@1001.service
- vboxadd-service.service
```

- Script Python d'audit local
- Vérification automatique de 9 contrôles

```
--- Utilisateurs avec shell interactif ---
root, tech, admin@tech

--- Contrôles de conformité ---
ssh_active : OK
ssh_custom_port : OK
ssh_port_listening : OK
root_login_disabled : OK
ufw_active : OK
fail2ban_installed : OK
fail2ban_active : OK
fail2ban_sshd_active : OK
password_auth_disabled : OK

Résumé : 9 / 9 contrôles validés
```

Exports en JSON et en CSV des résultats du script

1	key	value
2	audit_date	2026-04-15 10:38:17
3	hostname	tech
4	ssh_service_status	active
5	ssh_configured_port	2222
6	ssh_permit_root_login	no
7	ufw_status	actif
8	fail2ban_installed	True
9	fail2ban_status	active
10	open_ports	53, 68, 2222
11	active_services	cron.service, dbus.service, fail2ban.service, fwupd.service, getty@tty1.service, ModemManager.service, multipathd.service, polkit.service, rsyslog.service, ssh.service, systemd-journald.service, systemd-logind.service, systemd-networkd.service, systemd-resolved.service, systemd-udev.service, udisks2.service, unattended-upgrades.service, upower.service, user@1000.service, user@1001.service, vboxadd-service.service
12	interactive_shell_users	root, tech, admintech
13	validated_checks	9
14	total_checks	9
15	compliance_ssh_active	True
16	compliance_ssh_custom_port	True
17	compliance_ssh_port_listening	True
18	compliance_root_login_disabled	True
19	compliance_ufw_active	True
20	compliance_fail2ban_installed	True
21	compliance_fail2ban_active	True
22	compliance_fail2ban_sshd_active	True
23	compliance_password_auth_disabled	True
24		

```
resultats > cat audit_result.json > ...
1  {
2    "audit_date": "2026-04-15 10:38:17",
3    "hostname": "tech",
4    "ssh": {
5      "service_status": "active",
6      "configured_port": "2222",
7      "permit_root_login": "no",
8      "password_authentication": "no"
9    },
10   "security_tools": {
11     "ufw_status": "actif",
12     "fail2ban_installed": true,
13     "fail2ban_status": "active"
14   },
15   "system": {
16     "open_ports": [
17       "53",
18       "68",
19       "2222"
20     ],
21     "active_services": [
22       "cron.service",
23       "dbus.service",
24       "fail2ban.service",
25       "fwupd.service",
26       "getty@tty1.service",
27       "ModemManager.service",
28       "multipathd.service",
29       "polkit.service",
30       "rsyslog.service",
31       "ssh.service",
32       "systemd-journald.service",
33       "systemd-logind.service",
34       "systemd-networkd.service",
35       "systemd-resolved.service",
36       "systemd-udev.service",
37       "udisks2.service",
```

```
38       "unattended-upgrades.service",
39       "upower.service",
40       "user@1000.service",
41       "user@1001.service",
42       "vboxadd-service.service"
43     ],
44     "interactive_shell_users": [
45       "root",
46       "tech",
47       "admintech"
48     ]
49   },
50   "compliance": {
51     "ssh_active": true,
52     "ssh_custom_port": true,
53     "ssh_port_listening": true,
54     "root_login_disabled": true,
55     "ufw_active": true,
56     "fail2ban_installed": true,
57     "fail2ban_active": true,
58     "fail2ban_sshd_active": true,
59     "password_auth_disabled": true
60   },
61   "summary": {
62     "validated_checks": 9,
63     "total_checks": 9
64   }
65 }
```

Bilan de la mission

- Audit initial réalisé
- SSH durci
- UFW actif
- fail2ban actif et testé
- Logs consultés
- Script Python opérationnel

Recommandations pour TechSud

- Conserver l'authentification SSH par clé
- Centraliser les logs
- Mettre HTTPS sur les services web exposés

MERCI POUR VOTRE ATTENTION